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Features

Regulated Converters

- 1kVDC & 2kVDC Isolation
- UL94V-0 Package Material
- No Heatsink Required
- No External Components Required
- Toroidal Magnetics
- Optional Continuous Short Circuit Protected

Description

The RSZ series DC/DC converter has been designed for isolating or converting DC power rails where an SMD format with regulated output is required, although it is no larger than a standard unregulated SMD converter.

Selection Guide

Part Number SMD	2kVDC	Input Voltage (VDC)	Output Voltage (VDC)	Output Current (mA)	Max Capacitive Load[1]
Regulated without short circuit protection					
RSZ-xx05	(H)	3.3, 5, 12, 15, 24	5	200	1200µF
RSZ-xx09	(H)	3.3, 5, 12, 15, 24	9	111	1200µF
RSZ-xx12	(H)	3.3, 5, 12, 15, 24	12	42	680µF
Regulated with short circuit protection					
RSZ-xx3.3/P	(H)	3.3, 5, 12, 15, 24	3.3	200	3300µF
RSZ-xx05/P	(H)	3.3, 5, 12, 15, 24	5	200	1200µF

xx= Input Voltage (other input and output voltage combinations available on request
add suffix -R for tape & reel packing e.g. RSZ-0505-R

Specifications (measured at T_A = 25°C, nominal input voltage, full load and after warm-up)

Input Voltage Range			±5%
Output Voltage Accuracy			±2%
Line Voltage Regulation			1% max.
Load Voltage Regulation			1% max.
Output Ripple and Noise (at 20MHz BW)			100mVp-p max.
Operating Frequency			20kHz min. / 50kHz typ. / 85kHz max.
Efficiency at Full Load			50% min. / 60% typ.
Minimum Load			0%
No Load Power Consumption			134mW min. / 217mW typ. / 350mW max.
Isolation Voltage		(tested for 1 second)	1000VDC
		(rated for 1 minute)	500VAC / 60Hz
Isolation Voltage	H-Suffix	(tested for 1 second)	2000VDC
	H-Suffix	(rated for 1 minute)	1000VAC / 60Hz
Isolation Capacitance			70pF typ.
Isolation Resistance			10 GΩ min.
Short Circuit Protection			1 Second
P-Suffix			Continuous
Operating Temperature Range (free air)			-40°C to +70°C (see Graph)
Storage Temperature Range			-55°C to +125°C
Reflow Temperature		ROHS compliant	245°C (30 sec) max.
Vapour Phase Process			230°C (90 sec) max.
			(for more details see Application Notes)
Relative Humidity			95% RH
Package Weight			2.7g
P-Suffix			2.6g
Packing Quantity			33 pcs per tube
			500 pcs per Reel
MTBF (+25°C) } *For detailed information see Application Notes chapter "MTBF"*		RSZ types	1210 x 10^3 hours
		RSZ/P types	1688 x 10^3 hours
(+70°C)		RSZ types	149 x 10^3 hours
using MIL-HDBK 217F		RSZ/P types	418 x 10^3 hours

ECONOLINE
DC/DC-Converter
with 3 year Warranty

RECOM

1 Watt SMD Miniature Isolated Single Output



RoHS 2002/95/EC 5/6

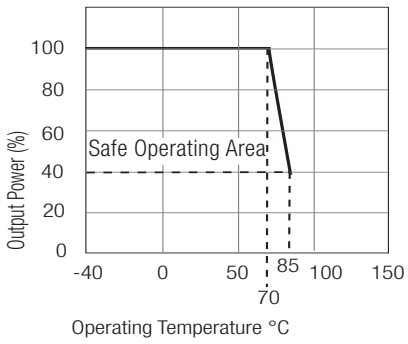
EN-60950-1 Certified

RSZ

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END OF LIFE: Please do not use for new design, replace with new SMD converter R1Z.

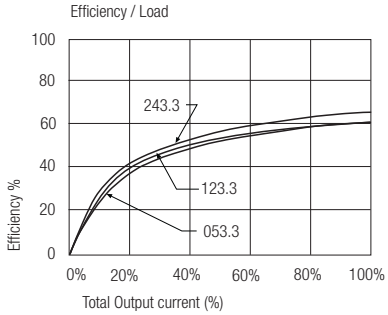
Derating-Graph
(Ambient Temperature)



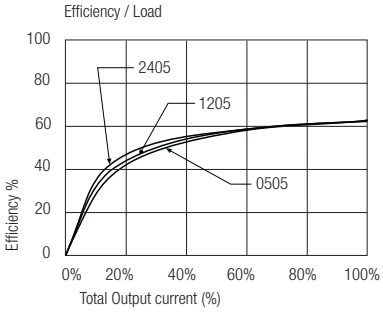
Refer to Application Notes

Typical Characteristics

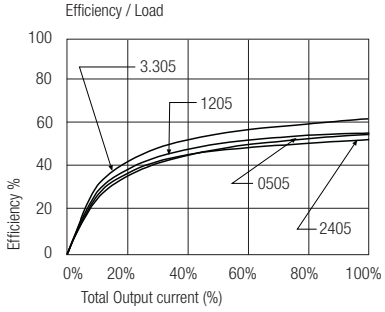
RSZ-xx3.3/P



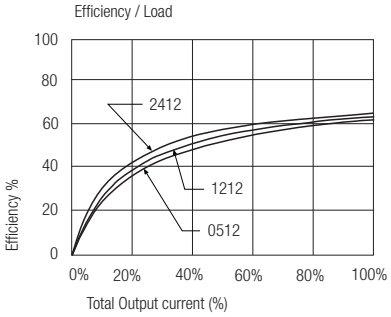
RSZ-xx05/P



RSZ-xx05



RSZ-xx12



Notes

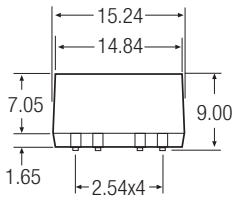
Note 1	Maximum capacitive load is defined as the capacitive load that will allow start up in under 1 second without damage to the converter

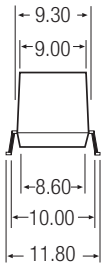
Package Style and Pinning (mm)

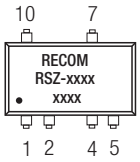
Ordering Example: RSZ-0505 (5V Input, 5V Output, not short circuit protected)
RSZ-0505/HP (5V Input, 5V Output, 2kVDC Isolation and short circuit protection)

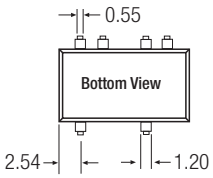
3rd angle projection

10 PIN Single SMD Package

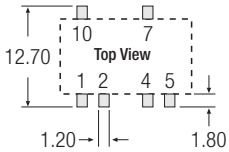








Recommended Footprint Details



Pin Connections

Pin #	Function
1	–Vin
2	+Vin
4	-Vout
5	-Vout
7	+Vout
10	NC

NC= No Connection

XX.X ± 0.5 mm
XX.XX ± 0.25 mm

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Mouser Electronics

Authorized Distributor

Click to View Pricing, Inventory, Delivery & Lifecycle Information:

RECOM:

RSZ-0505 RSZ-0505/H RSZ-0505/HP RSZ-0505/HP-R RSZ-0505/H-R RSZ-0505/P RSZ-0505/P-R RSZ-0505-R RSZ-0509 RSZ-0509/H RSZ-0509/H-R RSZ-0509-R RSZ-0512 RSZ-0512/H RSZ-0512/H-R RSZ-0512-R RSZ-053.3/HP RSZ-053.3/HP-R RSZ-053.3/P RSZ-053.3/P-R RSZ-1205 RSZ-1205/H RSZ-1205/HP RSZ-1205/HP-R RSZ-1205/H-R RSZ-1205/P RSZ-1205/P-R RSZ-1205-R RSZ-1209 RSZ-1209/H RSZ-1209/H-R RSZ-1209-R RSZ-1212 RSZ-1212/H RSZ-1212/H-R RSZ-1212-R RSZ-123.3/HP RSZ-123.3/HP-R RSZ-123.3/P RSZ-123.3/P-R RSZ-1505 RSZ-1505/H RSZ-1505/HP RSZ-1505/HP-R RSZ-1505/H-R RSZ-1505/P RSZ-1505/P-R RSZ-1505-R RSZ-1509 RSZ-1509/H RSZ-1509/H-R RSZ-1509-R RSZ-1512 RSZ-1512/H RSZ-1512/H-R RSZ-1512-R RSZ-153.3/HP RSZ-153.3/HP-R RSZ-153.3/P RSZ-153.3/P-R RSZ-2405 RSZ-2405/H RSZ-2405/HP RSZ-2405/HP-R RSZ-2405/H-R RSZ-2405/P RSZ-2405/P-R RSZ-2405-R RSZ-2409 RSZ-2409/H RSZ-2409/H-R RSZ-2409-R RSZ-2412 RSZ-2412/H RSZ-2412/H-R RSZ-2412-R RSZ-243.3/HP RSZ-243.3/HP-R RSZ-243.3/P RSZ-243.3/P-R RSZ-3.305 RSZ-3.305/H RSZ-3.305/HP RSZ-3.305/HP-R RSZ-3.305/H-R RSZ-3.305/P RSZ-3.305/P-R RSZ-3.305-R RSZ-3.309 RSZ-3.309/H RSZ-3.309/H-R RSZ-3.309-R RSZ-3.312 RSZ-3.312/H RSZ-3.312/H-R RSZ-3.312-R RSZ-3.33.3/HP RSZ-3.33.3/HP-R RSZ-3.33.3/P RSZ-3.33.3/P-R